

The Era of Printed Electronics: From 3D Printing to Inkjet Printing



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3D printing enthusiasts across the world have been looking forward to electronics printers.

In EFY's buyers' guide on 3D printers, published in March 2014 issue, Carine Carmy, director of marketing at Shapeways 3D Printing Marketplace and Community, was quoted as saying, "In the future, we are going to start seeing mixed materials, electronics 3D printing and conductive materials, as well." But then we had not imagined this would become a reality within a year!

Electronics printers are almost here

One of the interesting events at Consumer Electronics Show (CES) 2015 in Las Vegas, USA, was the launch of the pre-order campaign of Voxel8, which is claimed to be the first 3D electronics printer. Developed by Lewis Research Group at Harvard, under the guidance of Dr Jennifer A. Lewis, Voxel8 lets users blend plastic 3D printing material, an exclusive conductive paste and other electronic components into the same system to create highly-customisable objects from a simple antenna to a quadcopter.

The lab currently owns 17 patents (10 issued) on functional materials, print heads and other processes for 3D printing.

How it works

Voxel8 3D printer (Fig. 1) uses traditional fused deposition modelling/fused filament fabrication (FDM/FFF) technology along with conductive paste extrusion. It has two extruders (also known as print heads), of which one is the traditional FDM head that is seen in all 3D printers and which extrudes polylactic acid (PLA).

What makes this product special is the second print head that extrudes a special conductive ink made of highly-conductive silver particles developed at Lewis Re-

search Group lab at Harvard University. Also, the ink dries in about five minutes and can interconnect thin-quad-flat-pack (TQFP) integrated circuits (ICs).

Embedding components. The printer lets you insert or embed external electronic components such as ICs, resistors and capacitors into the design while it is being printed, and the printing process continues after this is done. This was achieved by partnering with Autodesk to create a specialised software that is hosted in the cloud and lets you design your product in 3D, where you can drag-and-drop electronic components, and add wires and connections to it.

During the printing process, the device



Fig. 1: Voxel8 3D electronics printer (Image courtesy: Voxel8)

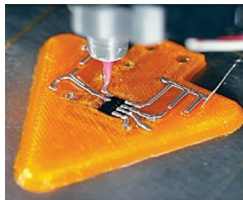


Fig. 2: Voxel8 lets users embed electronic components into their designs